FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES

CS054
(INTELLIGENT SYSTEMS)

EXERCISE
2

Sentiment Analysis

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Members:	**Name** / **Role** (,) (,) (,) (,)
Section:	AN42
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I. PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design and evaluate solutions for computing problems. [PO: e]

II. COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- To design an Intelligent System based on methods learned. [CLO: 3]

III. INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Demonstrate understanding of Sentiment Analysis.
- Design and develop a simple program about Sentiment Analysis using Python.

IV. BACKGROUND INFORMATION

Sentiment Analysis can assist us in determining the mood and feelings of the general public as well as obtaining useful information about the setting. Sentiment Analysis is the process of assessing data and categorizing it according to the research's needs.

Sentiments can be used to gain a better understanding of different events and their consequences. It goes by various names, including "sentiment analysis," "opinion mining," "opinion extraction," "sentiment mining," "subjectivity analysis," "affect analysis," "emotion analysis," and "review mining," but they all serve the same objective. We might be able to figure out what people like or want by evaluating their feelings.



TextBlob

TextBlob is a python library and offers a simple API to access its methods and perform basic NLP tasks. Textblob's sentiment function returns polarity and subjectivity as two characteristics.

The polarity ranges from [-1,1], with 1 denoting a positive statement and -1 denoting a negative one. Subjective statements usually refer to personal feelings, emotions, or judgments, whereas objective sentences refer to facts. Subjectivity is also a float with a value between 0 and 1.

V. GRADING SYSTEM / RUBRIC (please see separate sheet)

VI. LABORATORY ACTIVITY

Perform the following:

1. Add the following code:

```
from textblob import TextBlob
```

2. Declare a variable that will accept a sentence:

```
sentence = input("Enter a sentence : ")
```

3. Next initialize TextBlob

```
s = TextBlob(sentence)
```

4. Then check the polarity of the sentence by using the code below:

```
p = s.sentiment.polarity
```

5. Make a decision based on the polarity. If the polarity of the sentence is less than zero the sentiment is negative. If the polarity is equals to zero, then the sentiment is neutral. If the polarity is greater than zero, the sentiment is positive.

```
#Negative p<0, Neutral p=0, Positive p>0
if p <0:
    print("Negative")
elif p==0:
    print("Neutral")
elif p>0 and p<=1:
    print("Positive")
```

6. Make the program perform the sentiment analysis 10 times and make a summary of how many sentences are positive, neutral or negative.

OUTPUT:

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Dave> python -u "c:\Users\Dave\Desktop\TA5_Salvador.py"

Sentence 1
Enter a sentence: I love this beautiful day!
Polarity: 0.75
Sentiment: Positive

Sentence 2
Enter a sentence: I am very happy with my results.
Polarity: 1.0
Sentiment: Positive

Sentence 3
Enter a sentence: The food was delicious and amazing.
Polarity: 0.8
Sentiment: Positive

Sentence 4
Enter a sentence: I enjoyed spending time with my friends.
Polarity: 0.5
Sentiment: Positive

Sentence 5
Enter a sentence: Today is a normal day.
Polarity: 0.15
Sentiment: Positive
```

```
Sentence 6
Enter a sentence: The book is on the table.
Polarity: 0.0
Sentiment: Neutral

Sentence 7
Enter a sentence: I went to school this morning.
Polarity: 0.0
Sentiment: Neutral

Sentence 8
Enter a sentence: I hate waiting for a long time.
Polarity: -0.42500000000000004
Sentiment: Negative

Sentence 9
Enter a sentence: I feel sad and disappointed today.
Polarity: -0.625
Sentiment: Negative

Sentence 10
Enter a sentence: The movie was boring and terrible.
Polarity: -1.0
Sentiment: Negative

===== SENTIMENT ANALYSIS SUMMARY =====
Positive sentences: 5
Neutral sentences: 2
Negative sentences: 3
Total sentences analyzed: 10
PS C:\Users\Dave> python -u "c:\Users\Dave\Desktop\TA5_Salvador.py"
```

CODE:

```
from textblob import TextBlob

# Counters for each sentiment
positive = 0
neutral = 0
negative = 0

# Analyze 10 sentences
for i in range(10):
    print(f"\nSentence {i + 1}")
    sentence = input("Enter a sentence: ")

    # Initialize TextBlob
    s = TextBlob(sentence)

    # Get sentiment polarity
    p = s.sentiment.polarity

    print("Polarity:", p)

    # Determine sentiment
    if p < 0:
        print("Sentiment: Negative")
        negative += 1
    elif p == 0:
        print("Sentiment: Neutral")
        neutral += 1
    else:
        print("Sentiment: Positive")
        positive += 1

# Display summary
print("\n===== SENTIMENT ANALYSIS SUMMARY =====")
print("Positive sentences:", positive)
print("Neutral sentences:", neutral)
print("Negative sentences:", negative)
print("Total sentences analyzed:", positive + neutral + negative)
```

CODE SCREENSHOT:

TA5_Salvador.py

C: > Users > Dave > Desktop > TA5_Salvador.py > ...

```python
from textblob import TextBlob
# Counters for each sentiment
positive = 0
neutral = 0
negative = 0
# Analyze 10 sentences
for i in range(10):
    print(f"\nSentence {i + 1}")
    sentence = input("Enter a sentence: ")

    # Initialize TextBlob
    s = TextBlob(sentence)

    # Get sentiment polarity
    p = s.sentiment.polarity

    print("Polarity:", p)

    # Determine sentiment
    if p < 0:
        print("Sentiment: Negative")
        negative += 1
    elif p == 0:
        print("Sentiment: Neutral")
        neutral += 1
    else:
        print("Sentiment: Positive")
        positive += 1
# Display summary
print("\n===== SENTIMENT ANALYSIS SUMMARY =====")
print("Positive sentences:", positive)
print("Neutral sentences:", neutral)
print("Negative sentences:", negative)
print("Total sentences analyzed:", positive + neutral + negative)

```

VII. CHALLENGE

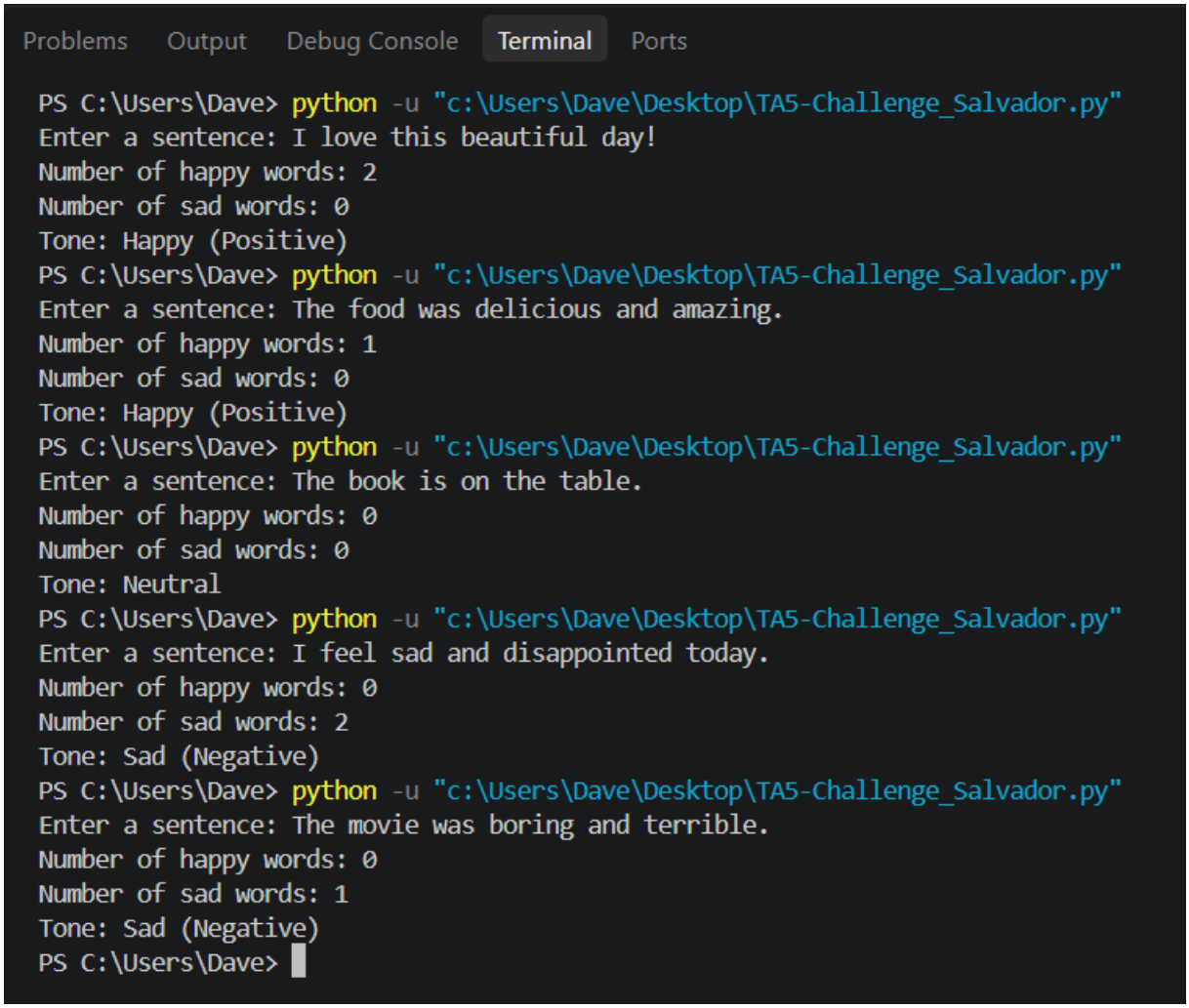
Write a program that will ask the user for a string containing a sentence. The program will then compute the happiness and sadness level of the sentence using the two functions described below. If the happiness level is higher than sadness level, then the tone of the sentence is happy. If the sadness level is higher, then the tone of the sentence is sad. Otherwise, it is neutral.

Find and print the tone of the sentence by printing the number of happy words followed by the number of sad words, followed by a simple statement of the analysis (whether positive, negative or neutral).

To accomplish this, you will write a function called number_happy(sentence) which returns the number of words in a given string called “sentence” that are happy. To do this, look for a public dataset containing a list of happy words. Find the total number of words in “sentence” that are in the happy words’ dataset.

Next, write a second function called number sad(sentence) that works the same way but instead counts the number of sad words, also from a separate dataset.

OUTPUT:


Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\Dave> python -u "c:\Users\Dave\Desktop\TA5-Challenge_Salvador.py"
Enter a sentence: I love this beautiful day!
Number of happy words: 2
Number of sad words: 0
Tone: Happy (Positive)
PS C:\Users\Dave> python -u "c:\Users\Dave\Desktop\TA5-Challenge_Salvador.py"
Enter a sentence: The food was delicious and amazing.
Number of happy words: 1
Number of sad words: 0
Tone: Happy (Positive)
PS C:\Users\Dave> python -u "c:\Users\Dave\Desktop\TA5-Challenge_Salvador.py"
Enter a sentence: The book is on the table.
Number of happy words: 0
Number of sad words: 0
Tone: Neutral
PS C:\Users\Dave> python -u "c:\Users\Dave\Desktop\TA5-Challenge_Salvador.py"
Enter a sentence: I feel sad and disappointed today.
Number of happy words: 0
Number of sad words: 2
Tone: Sad (Negative)
PS C:\Users\Dave> python -u "c:\Users\Dave\Desktop\TA5-Challenge_Salvador.py"
Enter a sentence: The movie was boring and terrible.
Number of happy words: 0
Number of sad words: 1
Tone: Sad (Negative)
PS C:\Users\Dave>

CODE:

```
import string

# List of happy and sad words
happy_words = [
    "happy", "joy", "love", "great", "wonderful",
    "excellent", "excited", "smile", "glad", "beautiful",
    "amazing", "delighted", "cheerful", "fantastic"
]

sad_words = [
    "sad", "cry", "hate", "bad", "terrible",
    "angry", "upset", "lonely", "unhappy", "hurt",
    "depressed", "miserable", "disappointed", "pain"
]

# Function to count happy words
def number_happy(sentence):
    words = sentence.lower().translate(
        str.maketrans("", "", string.punctuation)
    ).split()

    count = 0
    for word in words:
        if word in happy_words:
            count += 1

    return count

# Function to count sad words
def number_sad(sentence):
    words = sentence.lower().translate(
        str.maketrans("", "", string.punctuation)
    ).split()

    count = 0
    for word in words:
        if word in sad_words:
            count += 1

    return count

# Ask the user for a sentence
sentence = input("Enter a sentence: ")
```

```
# Compute happiness and sadness levels
happy = number_happy(sentence)
sad = number_sad(sentence)

# Display results
print("Number of happy words:", happy)
print("Number of sad words:", sad)

# Determine the tone
if happy > sad:
    print("Tone: Happy (Positive)")
elif sad > happy:
    print("Tone: Sad (Negative)")
else:
    print("Tone: Neutral")
```

CODE SCREENSHOT:

TA5_Salvador.py | TA5-Challenge_Salvador.py

C: > Users > Dave > Desktop > TA5-Challenge_Salvador.py > ...

```python
import string

# List of happy and sad words
happy_words = [
    "happy", "joy", "love", "great", "wonderful",
    "excellent", "excited", "smile", "glad", "beautiful",
    "amazing", "delighted", "cheerful", "fantastic"
]

sad_words = [
    "sad", "cry", "hate", "bad", "terrible",
    "angry", "upset", "lonely", "unhappy", "hurt",
    "depressed", "miserable", "disappointed", "pain"
]

# Function to count happy words
def number_happy(sentence):
    words = sentence.lower().translate(
        str.maketrans("", "", string.punctuation)
    ).split()

    count = 0
    for word in words:
        if word in happy_words:
            count += 1

    return count

```

TA5_Salvador.py | TA5-Challenge_Salvador.py

C: > Users > Dave > Desktop > TA5-Challenge_Salvador.py > ...

```
17  def number_happy(sentence):
27      return count
28
29  # Function to count sad words
30  def number_sad(sentence):
31      words = sentence.lower().translate(
32          str.maketrans("", "", string.punctuation)
33      ).split()
34
35      count = 0
36      for word in words:
37          if word in sad_words:
38              count += 1
39
40      return count
41
42  # Ask the user for a sentence
43  sentence = input("Enter a sentence: ")
44
45  # Compute happiness and sadness levels
46  happy = number_happy(sentence)
47  sad = number_sad(sentence)
48
49  # Display results
50  print("Number of happy words:", happy)
51  print("Number of sad words:", sad)
52
53  # Determine the tone
54  if happy > sad:
55      print("Tone: Happy (Positive)")
56  elif sad > happy:
57      print("Tone: Sad (Negative)")
58  else:
59      print("Tone: Neutral")
60
```

VIII. REFERENCES

- Artasanchez A. Joshi P.(2020). Artificial Intelligence with Python: You Complete Guide to Building Intelligent apps using Python 3.x 2nd Edition. Packt Publishing
- Attaway, Stormy (2013). MATLAB ®. A Practical Introduction to Programming and Problem Solving Third Edition. Elsevier, Inc.

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- https://research.google.com/colaboratory
- https://www.youtube.com/watch?v=RLYoEyIHL6A